

LLM Council v3.0 - Project Report

1. Project Overview & Architecture

LLM Council v3.0 is a monumental shift from the legacy FastAPI/LiteLLM architecture to a fully offline, production-ready local inference stack. The platform is designed to provide secure, local, and private deliberations via a council of 5 independent AI models.

Architecture Stack:

- Frontend: Next.js (App Router), React, Tailwind CSS, Framer Motion for premium animations.
- ML Inference Engine: Asynchronous Python microservice (aiohttp) paired with llama-cpp-python.
- Containerization: Fully Dockerized via docker-compose with seamless network orchestration.

2. Security & Privacy Levels

Security is absolute in v3.0:

- Offline Isolation: 100% of inference is executed locally using quantized GGUF models. No data is sent to external APIs (OpenAI/Anthropic).
- API Protection: The ML Engine employs strict CORS policies. The frontend sanitizes all queries before sending them via Server Actions or sanitized REST hooks.
- Dependency Risk: Removed obsolete, vulnerable libraries. Locked Next.js and Python dependencies.

3. Council Members & ML Models

The project automatically downloads heavily-optimized 1B to 3B parameter models for standard hardware:

1. Sage (Philosopher) -> Llama-3.2-1B-Instruct-Q4
2. Analyst (Logician) -> Qwen2.5-1.5B-Instruct-Q4
3. Strategist (Visionary) -> Phi-3-mini-4k-instruct-Q4
4. Skeptic (Challenger) -> Gemma-2-2B-it-Q4
5. Judge (Head) -> Llama-3.2-3B-Instruct-Q4